2025 Ch H2 Q9
Section: Chemistry in Society
Topic: Oxidising and Reducing Agents

Question Summary
Hydrogen peroxide forms hydroxyl radicals that start a chain reaction. Identify the missing step name and the meaning of •; state the term for enzyme shape change; give a consumer product with free■radical scavengers; and interpret an experiment on catalase activity vs pH following mass loss.

Worked Solution
(a)(i) The steps after initiation and before termination are **propagation** steps (radical reacts to form a new radical and continue the chain).

(a)(ii) The dot • denotes an **unpaired electron** (a free radical).

(b) When enzyme bonds are broken so the protein unfolds and changes shape, the enzyme is **denatured** (**denaturation**).

(c) Example of a consumer product with free■radical scavengers: **food products such as cooking oils/margarine** (antioxidants added), or **skin/face creams** (antioxidants). Either example gains full credit.

(d)(i) The mass decreases because **oxygen gas is produced and escapes** from the apparatus during the breakdown of H2O2.

(d)(ii) Two further variables to keep constant (any two):
- **Temperature** of the reaction mixture.
- **Size/thickness/surface area or mass** of each potato disc.

(Also acceptable: the **concentration of H_2O_2** used; total **volume** of solutions is already specified.)

(d)(iii) From the graph, the **rate is highest around pH 7** and lower at more acidic (pH 1, 4) and more alkaline (pH 10, 14) conditions. Thus, catalase activity in potato shows an **optimum near neutral pH**; moving away from neutral decreases the rate.

Final Answers

(a)(i) Propagation
(a)(ii) An unpaired electron (free radical)
(b) Denaturation
(c) Example: cooking oils/margarine with antioxidants (or antioxidant skin creams)
(d)(i) Loss of O_2 gas
(d)(ii) Keep temperature and potato disc size/surface area constant (any two valid controls)
(d)(iii) Rate is greatest near pH 7; lower at very acidic or very alkaline pH

Revision Tips

- Chain reactions: *initiation* $\rightarrow$ *propagation* $\rightarrow$ *termination*.
- Free■radical scavengers (antioxidants) interrupt propagation by reacting with radicals to form stable products.
- Enzymes have an optimal pH; extremes can denature the active site and slow the reaction.